List of Content

List of Image

Abstract

This project presents a secure IoT-based data transmission system that combines lightweight encryption, quantum-inspired key generation, and steganography to enhance data security. Environmental parameters such as temperature, humidity, and pressure are collected using sensors interfaced with an ESP32 microcontroller. The acquired data is converted into binary format and encrypted using an XOR-based technique to ensure confidentiality. A simulated Quantum Key Distribution (QKD) mechanism is employed to generate secure and dynamic encryption keys. In addition to encryption, the system uses Least Significant Bit (LSB) steganography to embed hidden information within images, providing an extra layer of security. The encrypted data and encoded images are transmitted over a wireless network to a server system. At the receiver end, the data is decrypted and reconstructed accurately, while hidden messages are extracted from images. A web-based dashboard is used to display real-time sensor data for monitoring purposes. The system demonstrates reliable performance with minimal computational overhead, making it suitable for resource-constrained IoT devices. Overall, the integration of encryption and steganography provides a multi-layered approach to secure communication.